

1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

ANS:- Comparison Table: DDR3 vs DDR4 vs DDR5 RAM

Feature	DDR3	DDR4	DDR5
Speed	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
Voltage	1.5 V	1.2 V	1.1 V
Power Consumption	High	Lower	Lowest and most efficient
Maximum Capacity (per module)	Up to 16 GB	Up to 64 GB	64 GB–256 GB (depending on module)
Performance	Good for older PCs	Excellent for most modern PCs	Best for gaming, AI, video editing, and workstations
Typical Use Cases	Older desktops/laptops, office work	Gaming, programming, multitasking	High-end gaming, AI, 3D rendering, professional content creation

2.

Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully. Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

ANS:-



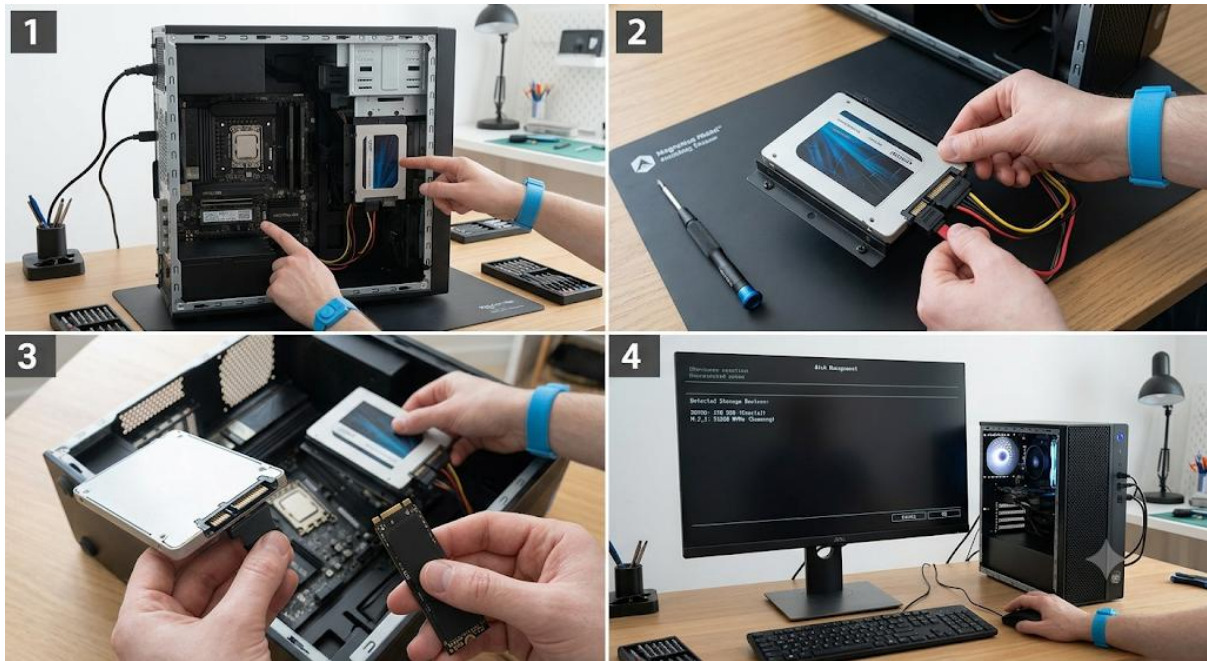
3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

ANS: -

<u>Features</u>	<u>HDD</u>	<u>SATA SSD</u>	<u>NVMe</u>
<u>Technology</u>	<u>Magnetic spinning disks</u>	<u>Flash memory</u>	<u>Flash memory over PCIe</u>
<u>Best for</u>	<u>Large file storage</u>	<u>Everyday computing</u>	<u>Gaming, programming, video editing</u>
<u>Speed</u>	<u>Slow</u>	<u>Fast</u>	<u>Very fast</u>
<u>Daily use case/app</u>	<u>Storing movies, photos, lecture videos, backups</u>	<u>Microsoft Office, Chrome, Visual Studio Code</u>	<u>GTA V, Valorant, Adobe Premiere Pro, Android Studio, AI/ML projects</u>

4.

Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly. Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.



Step 1: Locate: Open the desktop tower with the power disconnected and locate your storage drives (such as a 2.5-inch SATA SSD or an M.2 NVMe SSD on the motherboard).

Step 2: Remove: Disconnect the power and data cables, loosen any mounting screws, and take out the storage drive.

Step 3: Inspect & Reinstall: Identify the connector type (SATA or M.2/NVMe) and securely reinstall the drive back into its slot or bay.

Step 4: Verify: Reassemble the system, power it on, and check the BIOS or operating system to confirm the drive is recognized.